A Prime-Uniform Construction of Small Finite-Field Nikodym Sets

Abstract

We construct small Nikodym sets in $\mathbb{F}_q^h$. Let K be a fixed totally real field of degree d, and suppose that the rational prime q splits completely in K. We construct a product set whose complement is Nikodym and whose size is

$$\gg_{K,h} q^{h-2^{1-h}-(3h-5+2^{2-h})/d}.$$

In particular, the loss in the exponent is $O(h/d)$. We then use a finite family of multiquadratic fields such that every sufficiently large prime splits completely in at least one member. This yields a removed set of size $q^{h-2^{1-h}-o(1)}$ uniformly over the primes.

1 Statement and setup

We use the following convention. A set $S \subseteq \mathbb{F}_q^h$ is a *Nikodym set* if, for every $z \in \mathbb{F}_q^h$, there is an affine line L_z through z such that

$$L_z \setminus \{z\} \subseteq S.$$

Our fixed-field result is as follows.

Theorem 1.1 (Fixed totally real field)**.** *Let $h \geq 2$ be an integer, and let K be a totally real number field of degree d. Put*

$$\kappa_h = 3h - 5 + 2^{2-h}.$$

There is a constant $c_{K,h} > 0$ such that, whenever the rational prime q splits completely in K, there is a Nikodym set $S \subseteq \mathbb{F}_q^h$ satisfying

$$|S| \leq q^h - c_{K,h} q^{h-2^{1-h}-\kappa_h/d}. \tag{1}$$

In particular, since $\kappa_h < 4h$,

$$|S| \leq q^h - c_{K,h} q^{h-2^{1-h}-4h/d}. \tag{2}$$

Corollary 1.2 (Prime-uniform bound)**.** *Fix $h \geq 2$. For every $\varepsilon > 0$ and all sufficiently large primes q, there is a Nikodym set $S \subseteq \mathbb{F}_q^h$ such that*

$$|S| \leq q^h - q^{h-2^{1-h}-\varepsilon}. \tag{3}$$

Thus, uniformly over the primes, there are Nikodym sets whose complements have size at least $q^{h-2^{1-h}-o(1)}$.

The construction encodes the last coordinate using $h-1$ mixed-radix digits. We retain a large fiber on which the quadratic traces of all partial sums are fixed, and use the digit vector as the direction of the tangent line. A one-step rigidity lemma then recovers the highest digit at which two encodings could differ.

Next, let us define $\mathrm{IM}(h,r,q)$ to be the large induced matching in $\mathbb{F}_q^h$ between points and codimension r hyperplanes. We show the following bound.

Theorem 1.3 (Fixed totally real field)**.** *Let $h > r \geq 1$ be integers, and let K be a totally real number field of degree d.*

There is a constant $c_{K,h} > 0$ such that, whenever the rational prime q splits completely in K, we have

$$\mathrm{IM}(h,r,q) \geq c_{K,h} q^{h-(h-r)/2^r-(r+(h-r)^2)(1/d)}. \tag{4}$$

2 Sketch of ideas

In this section, we present some concepts underlying HPVZ in a more abstract framework. We shall then argue why they work.

Let us start with a simple example.

Lemma 2.1. *Let q be a prime. There exists a subset $P \subset \mathbb{F}_q^2$ of size $\Omega(q^{3/2})$, an assignment of slopes $p \mapsto s_p$ for $p \in P$, and a coloring $C : P \to [q]$, so that if $p, p' \in P$ satisfy $p' = p + h \cdot s_p$, then $C(p') = C(p) + h^2$.*

Consequently, given $S \subset [q]$ so that $S - S$ lacks non-zero squares, we get $\mathrm{IM}(2,q) \geq \Omega(|S|q^{1/2})$.

Proof. Let us start by building something over $\mathbb{Z}^2$. Namely, for each point $p = (a,b)$, we assign $C_0(p) := a + b^2$ and $s_p^{(0)} := (-2b, 1)$. If $p' = p + h \cdot s_p^{(0)}$, then we have that $C_0(p') = C_0(p) + h^2$. Indeed, writing $p = (a,b), p' = (a', b')$, we get $h = b' - b$, and then $C_0(p') - C_0(p) = -h \cdot 2b + (b + h)^2 - b^2$.

Now, let $P_0 := [q/2] \times [q^{1/2}/2]$. We claim that if $p, p' \in P_0$, and $p' \equiv p + h \cdot s_p \pmod q$ (for $|h| < q$), then $p' = p + h \cdot s_p$. Indeed, comparing second coordinates, we get that $|h| < q^{1/2}/2$. Whence, $||h \cdot s_p||_\infty \leq q/2$, and thus $||p' - p||_\infty < q/2$ implies that $p' = p + h \cdot s_p$.

So now we just take the obvious embedding $\iota : \mathbb{F}_q^2 \to [q]^2$.

The last estimate easily follows due to averaging. Start with $S \subset [q]$. Now pick $t \in \{-q, \ldots, q\}$ uniformly at random. For $p \in P$, and $s \in S$, $\Pr[s + t = C(p)] \geq 1/3q$. Thus $[\#(p \in P : \iota(p) \in t + S)] = |S||P|/3q = \Omega(|S|q^{1/2})$.

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Before giving the formal details, let us paint a quick picture.

Lemma 2.2. *Let q be a prime. There exists a subset $P \subset \mathbb{F}_q^3$ of size $\Omega(q^{3-1/4})$, an assignment of slopes $p \mapsto s_p$ for $p \in P$, a pair of coloring colorings $C_1, C_2 : P \to [q]$, so that if $p, p' \in P$ satisfy $p' = p + h \cdot s_p$, then $C_i(p') = C_i(p) + h^2$ for some $i \in \{1, 2\}$.*

Consequently, given $S \subset [q]$ so that $S - S$ lacks non-zero squares, we get $\mathrm{IM}(3,q) \geq \Omega(|S|^2 q^{3/4})$.

Proof. Pick auxiliary distinct primes $\frac{p^{1/2}}{100} \leq p_1, p_2 \leq \frac{p^{1/2}}{10}$.

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3 Preliminary lemmas

Let $\sigma_1, \ldots, \sigma_d : K \hookrightarrow \mathbb{R}$ be the real embeddings. For $T > 0$, put

$$\Lambda_T = \{x \in \mathcal{O}_K : |\sigma_j(x)| \le T \text{ for } 1 \le j \le d\}.$$

Under the Minkowski embedding, $\mathcal{O}_K$ is a full-rank lattice in $\mathbb{R}^d$. Consequently,

$$|\Lambda_T| \asymp_K T^d \tag{5}$$

as $T \to \infty$. We shall only use the lower bound.

Lemma 3.1 (Small elements do not vanish modulo $\mathfrak{q}$)**.** *Suppose that q splits completely in K, and choose a prime ideal $\mathfrak{q} \mid q$. Then $\mathcal{O}_K/\mathfrak{q} \cong \mathbb{F}_q$. If $0 \ne z \in \mathfrak{q}$, then*

$$|\operatorname{N}_{K/\mathbb{Q}}(z)| \ge q.$$

In particular, if $z \in \mathfrak{q}$ and $|\sigma_j(z)| < q^{1/d}$ for every j, then $z = 0$.

Proof. The ideal $\mathfrak{q}$ has norm q. If $0 \ne z \in \mathfrak{q}$, then $(z) \subseteq \mathfrak{q}$, so $(z) = \mathfrak{q}\mathfrak{a}$ for some integral ideal $\mathfrak{a}$. Hence

$$|\operatorname{N}_{K/\mathbb{Q}}(z)| = q \operatorname{N}(\mathfrak{a}) \ge q.$$

The final assertion follows by taking the product over the real embeddings. □

Lemma 3.2 (A product-complement criterion)**.** *Let $P = P_1 \times \cdots \times P_h \subseteq \mathbb{F}_q^h$, where $h \ge 2$. Suppose that for every $u \in P$ there is a line L_u through u such that $L_u \cap P = \{u\}$. Then $\mathbb{F}_q^h \setminus P$ is a Nikodym set.*

Proof. For $u \in P$, the assumed line satisfies $L_u \setminus \{u\} \subseteq \mathbb{F}_q^h \setminus P$. Now let $u \notin P$. Some coordinate, say u_j, does not belong to P_j. Choose $k \ne j$. The coordinate line $u + \mathbb{F}_q e_k$ has jth coordinate constantly equal to u_j, and is therefore disjoint from P. □

In addition to the d-dimensional geometry given to us by our number fields, it is also natural to sajkdkla.

Lemma 3.3 (Mixed-digit representation)**.** *Let $m \ge 1$, let $R_1, \ldots, R_m$ be positive integers, and let $0 < \rho < 1/2$. Define*

$$D_1 = 1, \qquad D_i = \prod_{j<i} R_j \quad (2 \le i \le m).$$

Then the map

$$\prod_{i=1}^{m} \Lambda_{\rho R_i} \longrightarrow \mathcal{O}_K, \qquad (w_1, \ldots, w_m) \longmapsto \sum_{i=1}^{m} D_i w_i,$$

is injective.

Proof. Suppose that two digit vectors have the same image, and put $\beta_i = w_i - w_i'$. Reducing $\sum_i D_i \beta_i = 0$ modulo $R_1\mathcal{O}_K$ gives $\beta_1 \in R_1\mathcal{O}_K$. If $\beta_1 \ne 0$, then

$$|\operatorname{N}_{K/\mathbb{Q}}(\beta_1)| \le (2\rho R_1)^d < R_1^d.$$

On the other hand, writing $\beta_1 = R_1\gamma$ with $0 \ne \gamma \in \mathcal{O}_K$ gives $|\operatorname{N}_{K/\mathbb{Q}}(\beta_1)| = R_1^d |\operatorname{N}_{K/\mathbb{Q}}(\gamma)| \ge R_1^d$, a contradiction. Thus $\beta_1 = 0$. Dividing by R_1 and repeating proves that $\beta_i = 0$ for every i. □

Because K is totally real,

$$\|x\| := \left(\frac{\operatorname{Tr}_{K/\mathbb{Q}}(x^2)}{d}\right)^{1/2} = \left(\frac{1}{d}\sum_{j=1}^{d}\sigma_j(x)^2\right)^{1/2}$$

defines a norm on $K \otimes_{\mathbb{Q}} \mathbb{R}$. Moreover, if $0 \neq x \in \mathcal{O}_K$, then the arithmetic–geometric mean inequality gives

$$\|x\|^2 \geq \left(\prod_{j=1}^{d}\sigma_j(x)^2\right)^{1/d} = |\operatorname{N}_{K/\mathbb{Q}}(x)|^{2/d} \geq 1. \tag{6}$$

Generally, the key geometric input underlying our construction is the following.

Proposition 3.4 ('Spheres and orthogonality'). *Let $Q, t \geq 1$ be integers. Suppose that $u, u' \in \Lambda^t_{Q/100t}$ and $w, w' \in \mathcal{O}_K^t$, and put*

$$x = u + Qw, \qquad x' = u' + Qw'.$$

If

$$\sum_{i=1}^{t}\operatorname{Tr}_{K/\mathbb{Q}}(x_i^2) = \sum_{i=1}^{t}\operatorname{Tr}_{K/\mathbb{Q}}((x_i')^2), \qquad \sum_{i=1}^{t}\operatorname{Tr}_{K/\mathbb{Q}}(w_i(x_i' - x_i)) = 0,$$

then $w = w'$.

Remark 3.5. Roughly speaking, one should intuitively expect this to hold due to the following heuristics. Firstly, the effect of u and u' should each be negligible, due to the scale of Q (if you imagine $\sum_i ||x_i||$ as morally being the Euclidean norm, this is a byproduct of 'continuity'). Thus, ignoring the small noise, the second condition translates to 'orthogonality' ($\langle w, w' - w\rangle = 0$). Now, adding two non-zero, orthogonal vectors together increases each of their norms (in Euclidean space), so we expect the two equalities to imply $w = w'$.

Proof. Set

$$v = x' - Qw = u' + Q(w' - w).$$

Since $u = x - Qw$, the assumptions imply

$$\begin{aligned}\sum_{i=1}^{t}\operatorname{Tr}_{K/\mathbb{Q}}(v_i^2 - u_i^2) &= \sum_{i=1}^{t}\operatorname{Tr}_{K/\mathbb{Q}}((x_i' - Qw_i)^2 - (x_i - Qw_i)^2) \\ &= \sum_{i=1}^{t}\operatorname{Tr}_{K/\mathbb{Q}}((x_i')^2 - x_i^2) - 2Q\operatorname{Tr}_{K/\mathbb{Q}}(w_i(x_i' - x_i)) = 0.\end{aligned}$$

Consequently,

$$\|u' + Q(w' - w)\| = \|v\| = \|u\|.$$

The triangle inequality now yields

$$Q\sum_{i=1}^{t}\|w_i' - w_i\| \leq \sum_{i=1}^{t}\|u_i' + Q(w_i' - w_i)\| + \|u_i\| \leq \frac{Q}{50}.$$

By (6), this is possible only if $w_i' = w_i$ for all $i \in [t]$. $\square$

4 REVAMP

Let us start by deriving the $r=1$ bound.perhaps not exactly the Theorem, since we do not obtain a product set. perhaps state as its own result, and pre-face the general sphere statement

Proof of Theorem 1.3 for $r=1$. Let $M=q^{1/d}$. Now take $A=\Lambda_{M/100}, B=\Lambda_{M^{1/2}/100h}$.

Color $p=(a,b_1,\dots,b_{h-1})\in A\times B^{h-1}$ by $\Phi(p):=\mathrm{Tr}_{K/\mathbb{Q}}(2a-\sum_{i=1}^{h-1}b_i^2)$. This takes at most $M/2$ values, thus we can pigeonhole to find $P\subset A\times B^{h-1}$ of size $\Omega_{K,h}(q^{(h+1)/2-1/d})$ which is monochromatic under Φ.

Now, for each $p\in P$, we assign the plane π_p defined by $X_0-\sum_{i=1}^{h-1}b_iX_i=a-\sum_{i=1}^{h-i}b_i^2=:\psi(p)$. Clearly $p\in\pi_p$. Let us first show that $((p,\pi_p))_{p\in P}$ is an induced matching in $\mathcal{O}_K^h$, afterwards we shall verify this is preserved under projection.

Suppose $p'\in P\cap\pi_p$ for some $p'\neq p$, and write $p'=(a',b_1',\dots,b_{h-1}')$. Then, we have that

$$\delta=(a'-a)-\sum_{i=1}^{h-1}b_i(b_i'-b_i)=\psi(p)-\psi(p)=0.$$

Next by definition of P,

$$0=\Phi(p)-\Phi(p')=\mathrm{Tr}_{K/\mathbb{Q}}(2(a'-a)-\sum_{i=1}^{h-1}(b_i'^2-b_i^2))=\mathrm{Tr}_{K/\mathbb{Q}}(2\delta)-\mathrm{Tr}_{K/\mathbb{Q}}(\sum_{i=1}^{h-1}(b_i'-b_i)^2).$$

Now, $\mathrm{Tr}_{K/\mathbb{Q}}(2\delta)=2\,\mathrm{Tr}_{K/\mathbb{Q}}(\delta)=0$, thus we get that $\sum_{i=1}^{h-1}\mathrm{Tr}_{K/\mathbb{Q}}((b_i'-b_i)^2)=0$, but this means $b_i'=b_i$ for all i (because non-zero squares have positive trace), whence $p'=p$ (since we get $a'-a=\delta$).

Now suppose $\mathfrak{p}\mid q$, and let $\varphi:\mathcal{O}_K/\mathfrak{p}\to\mathbb{F}_q$ be a homomorphism (and extend it coordinate-wise). We of course take $((\varphi(p),\varphi(\pi_p))_{p\in P}$. Note that $P\subset\Lambda_{M/4}^h$, thus $P-P\cap\ker(\varphi)=\{\vec{0}\}$, so this map is injective. Now suppose that $\varphi(p')\in\varphi(\pi_p)$ for some $p\in P$. We now get

$$\delta=a'-a-\sum_{i=1}^{h-1}b_i(b_i'-b_i)\in\mathfrak{p}.$$

But, as $P\subset A\times B^{h-1}$, we get that $\delta\in\Lambda_{M/2}^h$, and thus again $\delta=0$. So $p'\neq p$ would contradict that we have an induced matching over $\mathcal{O}_K^h$. $\square$

Remark 4.1. If we instead pigeonhole on $(\mathrm{Tr}_{K/\mathbb{Q}}(a),\mathrm{Tr}_{K/\mathbb{Q}}(b_1^2),\dots,\mathrm{Tr}_{K/\mathbb{Q}}(b_{h-1}^2))$, we get a product set of size $\Omega_{K,h}(q^{(h+1)/2-h/d})$.

As r increases, we manage to do better. The key that each equation we add allows us to achieve a better homomorphism. Or more accurately, we produce a set of maps, so that at least one of them behaves like a homomorphism.

5 The fixed-field construction

Proof of Theorem 1.1. It is enough to treat all sufficiently large q; the constant can be reduced at the end to cover the finitely many remaining split primes. Put

$$M=q^{1/d},\qquad\rho=\frac{1}{100h},\qquad\gamma=\frac{1}{10}.$$

For $1 \le i \le h-1$, define

$$Q_i = \left\lfloor M^{2^{-i}} \right\rfloor, \qquad W_i = \Lambda_{\rho Q_i}.$$

For $1 \le i \le h$, also define

$$D_i = \prod_{j<i} Q_j,$$

where the empty product is 1. We may assume that every $Q_i \ge 2$. Notice that

$$D_{i+1} = D_i Q_i \le M^{1-2^{-i}}, \qquad D_i Q_i^2 \le M. \tag{7}$$

Define

$$Y = \left\{ \sum_{i=1}^{h-1} D_i w_i : w_i \in W_i \right\}.$$

Lemma 3.3 shows that every element of Y has a unique digit representation. Since $Q_i \asymp_h M^{2^{-i}}$, the lattice estimate (5) gives

$$|Y| = \prod_{i=1}^{h-1} |W_i| \gg_{K,h} \prod_{i=1}^{h-1} Q_i^d \gg_{K,h} q^{1-2^{1-h}}. \tag{8}$$

For $y = \sum_i D_i w_i \in Y$, let

$$y_0 = 0, \qquad y_i = \sum_{j=1}^{i} D_j w_j \quad (1 \le i \le h-1).$$

For every real embedding σ,

$$|\sigma(y_i)| \le \rho \sum_{j=1}^{i} D_j Q_j = \rho \sum_{j=1}^{i} D_{j+1} \le \rho i D_{i+1}. \tag{9}$$

Color y by its energy vector

$$c(y) = (\mathrm{Tr}_{K/\mathbb{Q}}(y_1^2), \ldots, \mathrm{Tr}_{K/\mathbb{Q}}(y_{h-1}^2)).$$

Each entry is a nonnegative integer, and (9) shows that the ith entry takes $O_{K,h}(D_{i+1}^2)$ possible values. Hence the number of colors is

$$\begin{aligned} O_{K,h}\left(\prod_{i=1}^{h-1} D_{i+1}^2\right) &\le O_{K,h}\left(M^{2\sum_{i=1}^{h-1}(1-2^{-i})}\right) \\ &= O_{K,h}\left(M^{2(h-2+2^{1-h})}\right). \end{aligned}$$

By pigeonholing, there is a fiber $B \subseteq Y$ such that

$$|B| \gg_{K,h} q^{\,1-2^{1-h}-2(h-2+2^{1-h})/d}. \tag{10}$$

We also choose a large trace fiber. Since $\mathrm{Tr}_{K/\mathbb{Q}}(a)$ is an integer of absolute value at most $d\gamma M$ for $a \in \Lambda_{\gamma M}$, another pigeonhole argument gives an integer s and a set

$$A = \{a \in \Lambda_{\gamma M} : \mathrm{Tr}_{K/\mathbb{Q}}(a) = s\}, \qquad |A| \gg_{K,h} M^{d-1} = q^{1-1/d}. \tag{11}$$

Choose a prime ideal $\mathfrak{q} \mid q$, and let

$$\varphi : \mathcal{O}_K \longrightarrow \mathcal{O}_K/\mathfrak{q} \cong \mathbb{F}_q$$

be the reduction map. Lemma 3.1 shows that φ is injective on A: the difference of two elements of A has every embedding bounded by $2\gamma M < M$. It is also injective on Y. Indeed, (9) and (7) give

$$|\sigma(y)| \leq \rho(h-1)D_h < M/100 \qquad (y \in Y),$$

so the difference of two elements of Y has every embedding bounded by $M/50 < M$.

Set

$$P = \varphi(A)^{h-1} \times \varphi(B) \subseteq \mathbb{F}_q^h. \tag{12}$$

We construct a line tangent to P at each of its points. Let

$$p = (\varphi(a_1), \ldots, \varphi(a_{h-1}), \varphi(b)) \in P, \qquad b = \sum_{i=1}^{h-1} D_i w_i,$$

where the digit representation of b is unique, and define

$$L_p = \{p + \lambda(\varphi(w_1), \ldots, \varphi(w_{h-1}), 1) : \lambda \in \mathbb{F}_q\}. \tag{13}$$

Suppose that another point

$$p' = (\varphi(a'_1), \ldots, \varphi(a'_{h-1}), \varphi(b')) \in P$$

lies on L_p, and write $b' = \sum_i D_i w'_i$. If $b' = b$, then the last coordinate in (13) gives $\lambda = 0$, and hence $p' = p$. We may therefore suppose that $b' \neq b$, and choose the largest i for which $w'_i \neq w_i$. Put

$$t = b' - b.$$

By maximality of i, we have $t = y'_i - y_i$. The last coordinate of the line gives $\lambda = \varphi(t)$, while its ith coordinate gives

$$\delta := a'_i - a_i - w_i t \in \mathfrak{q}. \tag{14}$$

We claim that this congruence lifts to an equality in $\mathcal{O}_K$. By (9),

$$|\sigma(t)| \leq 2\rho i D_{i+1} \leq 2\rho h D_i Q_i.$$

Together with $|\sigma(w_i)| \leq \rho Q_i$ and (7), this gives

$$|\sigma(w_i t)| \leq 2\rho^2 h D_i Q_i^2 \leq 2\rho^2 h M.$$

Consequently,

$$|\sigma(\delta)| \leq (2\gamma + 2\rho^2 h)M < M$$

for every embedding σ. Lemma 3.1 applied to (14) therefore yields

$$a'_i - a_i = w_i t. \tag{15}$$

Since A is a trace fiber,

$$\mathrm{Tr}_{K/\mathbb{Q}}(w_i t) = 0. \tag{16}$$

We now apply Proposition 3.4. Set

$$u = y_{i-1}, \qquad u' = y'_{i-1}, \qquad x = y_i = u + D_i w_i, \qquad x' = y'_i = u' + D_i w'_i.$$

The prefix estimate gives

$$u, u' \in \Lambda_{D_i/100},$$

because $\rho(i-1) < 1/100$. Since b, b' lie in the same energy fiber,

$$\mathrm{Tr}_{K/\mathbb{Q}}(u^2) = \mathrm{Tr}_{K/\mathbb{Q}}((u')^2), \qquad \mathrm{Tr}_{K/\mathbb{Q}}(x^2) = \mathrm{Tr}_{K/\mathbb{Q}}((x')^2);$$

the first equality is automatic when $i = 1$. Finally, (16) says that $\mathrm{Tr}_{K/\mathbb{Q}}(w_i(x' - x)) = 0$. The proposition, with $Q = D_i$, forces $w_i = w'_i$, contradicting the choice of i.

Thus $L_p \cap P = \{p\}$ for every $p \in P$. By Lemma 3.2, the set $S = \mathbb{F}_q^h \setminus P$ is Nikodym. Finally, (10) and (11) give

$$|P| = |A|^{h-1}|B| \gg_{K,h} q^{h - 2^{1-h} - (3h-5+2^{2-h})/d}.$$

This proves (1), and hence (2), for all sufficiently large split primes q. For any remaining split prime, the complement of a one-point product set is Nikodym by Lemma 3.2; decreasing $c_{K,h}$ covers those finitely many primes as well. □

Remark 5.1. When $h = 2$, the exponent in (1) is $3/2 - 2/d$.

6 Extending to hyperplanes

Proof. Proof of Theorem 1.3 for $r = 1$ Let $M = q^{1/d}$. Now take $A = \Lambda_{M/100}, B = \Lambda_{M^{1/2}/100h}$.

Color $p = (a, b_1, \ldots, b_{h-1}) \in A \times B^{h-1}$ by $\Phi(p) := \mathrm{Tr}_{K/\mathbb{Q}}(2a - \sum_{i=1}^{h-1} b_i^2)$. This takes at most $M/2$ values, thus we can pigeonhole to find $P \subset A \times B^{h-1}$ of size $\Omega_{K,h}(q^{(h+1)/2 - 1/d})$ which is monochromatic under Φ.

Now, for each $p \in P$, we assign the plane π_p defined by $X_0 - \sum_{i=1}^{h-1} b_i X_i = a - \sum_{i=1}^{h-i} b_i^2 =: \psi(p)$. Clearly $p \in \pi_p$. Let us first show that $((p, \pi_p))_{p \in P}$ is an induced matching in $\mathcal{O}_K^h$, afterwards we shall verify this is preserved under projection.

Suppose $p' \in P \cap \pi_p$ for some $p' \neq p$, and write $p' = (a', b'_1, \ldots, b'_{h-1})$. Then, we have that

$$\delta = (a' - a) - \sum_{i=1}^{h-1} b_i(b'_i - b_i) = \psi(p) - \psi(p) = 0.$$

Next by definition of P,

$$0 = \Phi(p) - \Phi(p') = \mathrm{Tr}_{K/\mathbb{Q}}(2(a' - a) - \sum_{i=1}^{h-1}(b_i'^2 - b_i^2)) = \mathrm{Tr}_{K/\mathbb{Q}}(2\delta) - \mathrm{Tr}_{K/\mathbb{Q}}(\sum_{i=1}^{h-1}(b'_i - b_i)^2).$$

Now, $\mathrm{Tr}_{K/\mathbb{Q}}(2\delta) = 2\,\mathrm{Tr}_{K/\mathbb{Q}}(\delta) = 0$, thus we get that $\sum_{i=1}^{h-1} \mathrm{Tr}_{K/\mathbb{Q}}((b'_i - b_i)^2) = 0$, but this means $b'_i = b_i$ for all i (because non-zero squares have positive trace), whence $p' = p$ (since we get $a' - a = \delta$).

Now suppose $\mathfrak{p} \mid q$, and let $\varphi : \mathcal{O}_K/\mathfrak{p} \to \mathbb{F}_q$ be a homomorphism (and extend it coordinate-wise). We of course take $((\varphi(p), \varphi(\pi_p))_{p \in P}$. Note that $P \subset \Lambda_{M/4}^h$, thus $P - P \cap \ker(\varphi) = \{\vec{0}\}$, so this map is injective. Now suppose that $\varphi(p') \in \varphi(\pi_p)$ for some $p \in P$. We now get

$$\delta = a' - a - \sum_{i=1}^{h-1} b_i(b'_i - b_i) \in \mathfrak{p}.$$

But, as $P \subset A \times B^{h-1}$, we get that $\delta \in \Lambda_{M/2}^h$, and thus again $\delta = 0$. So $p' \neq p$ would contradict that we have an induced matching over $\mathcal{O}_K^h$. $\square$

Remark 6.1. If we instead pigeonhole on $(\mathrm{Tr}_{K/\mathbb{Q}}(a), \mathrm{Tr}_{K/\mathbb{Q}}(b_1), \ldots, \mathrm{Tr}_{K/\mathbb{Q}}(b_{h-1}))$, we get a product set of size $\Omega_{K,h}(q^{(h+1)/2-h/d})$.

We shall now proceed to the general case.

Proof. Set $t := h - r$. Set $M = q^{1/d}, \rho := 1/100h$. For $1 \leq i \leq r$, set

$$Q_i := \lfloor M^{2^{-i}} \rfloor, W_i := \Lambda_{\rho Q_i},$$

also set $D_1 := 1$ and $D_{i+1} := Q_i \cdot D_i$ for $i \in [t]$.

We define $B := \sum_{i=1}^r D_i w_i : w \in W_i\}$. Lemma 3.3 implies that $|Y| = \prod_{i=1}^r |W_i| \gg_{K,h} D_{r+1}^d = q^{1-2^{-r}}$. Meanwhile, we take $A := \Lambda_{\rho M}$. We shall define a coloring χ on $A^r \times B^t$ with $M^{O_{h,r}(1)}$ colors; to do so we must introduce some further notation.

Specifically, given $b = \sum_i D_i w_i$, let

$$b^{(0)} = 0, \quad b^{(i)} := \sum_{j=1}^{i} D_i w_j \quad i \in [r].$$

We now assign[1]

$$\chi(b) := (||b^{(0)}||, ||b^{(1)}||, \ldots, ||b^{(r)}||).$$

Now, we clearly have that c takes most $(h(\rho M)^2)^r \leq M^{2r}$ values (since $||b^{(0)}|| = 0$). Thus we may take $B' \subset B$ of size $|B'| \gg_{K,h} q^{1-2^{-r}-2r/d}$.

We also pass to $A' \subset A$ where $\mathrm{Tr}_{K/\mathbb{Q}}(a)$ is constant, thus $|A'| \gg_{K,h} q^{1-1/d}$ now. We set $P := A'^r \times B'^t$. Now, given $p = (a_1, \ldots, a_r, b_1, \ldots, b_t) \in P$, writing $b_i = \sum_{j=1}^r D_j w_{i,j}$, we define π_p to be the solution to the equations

$$X_j - \sum_{i=1}^{t} w_{i,j} Y_i = a_j - \sum_{i=1}^{t} w_{i,j} b_i,$$

for $j \in [r]$. Clearly $p \in \pi_p$. Now, let us verify this is an induced matching.

Suppose $p' \in \pi_p$, but $p' \neq p$. Then there must be some maximal $j \in [r]$ where there exists $i \in [t]$ so that $w'_{i,j} \neq w_{i,j}$. Set

$$\delta_j := a'_j - a_j - \sum_{k=1}^{t} w_{k,j}(b'_k - b_k) = 0.$$

We have that

$$0 = \mathrm{Tr}_{K/\mathbb{Q}}(2(a'_j - a_j)) - \mathrm{Tr}_{K/\mathbb{Q}}($$

Note that $b'_k - b_k = b_k^{(j)\prime} - b_k^{(j)} = b_k^{(j-1)\prime} - b_k^{(j-1)} + D_j(w'_{k,j} - w_{k,j})$. Consequently, as $||b_k^{(j-1)}|| = ||b_k^{(j-1)\prime}||, ||b_k^{(j-1)} + D_j w_{k,j}|| = ||b_k^{(j-1)\prime} + D_j w'_{k,j}||$ for all k, we get that $w_{k,j} = w'_{k,j}$, contradiction. $\square$

[1]Recall that $||x|| := \sqrt{\mathrm{Tr}_{K/\mathbb{Q}}(x^2)}$.

7 A finite split family and the prime-uniform bound

Proof of Corollary 1.2. Fix $m \geq 1$. Choose $2m$ distinct odd rational primes, grouped into pairs (ℓ_j, ℓ_j') for $1 \leq j \leq m$, and consider the finite family

$$\mathcal{K}_m = \left\{ \mathbb{Q}(\sqrt{r_1}, \ldots, \sqrt{r_m}) : r_j \in \{\ell_j, \ell_j', \ell_j\ell_j'\} \right\}. \tag{17}$$

Every field in this family is totally real and has degree 2^m. Indeed, the squareclasses of the r_j are independent: they are nontrivial and have supports in pairwise disjoint pairs of primes.

Let q be an odd prime distinct from those used in (17). For each j, the two Legendre symbols

$$\left(\frac{\ell_j}{q}\right), \qquad \left(\frac{\ell_j'}{q}\right)$$

belong to $\{\pm 1\}$. Among these two signs and their product, at least one equals $+1$. We may therefore choose $r_j \in \{\ell_j, \ell_j', \ell_j\ell_j'\}$ such that

$$\left(\frac{r_j}{q}\right) = 1.$$

Then q splits completely in every $\mathbb{Q}(\sqrt{r_j})$, and hence splits completely in their compositum $K \in \mathcal{K}_m$.

The family $\mathcal{K}_m$ is finite. Taking the minimum of the constants $c_{K,h}$ supplied by Theorem 1.1 over $K \in \mathcal{K}_m$, and increasing the threshold for q if necessary, gives a constant $c_{m,h} > 0$ such that every sufficiently large prime q admits a Nikodym set satisfying

$$|S| \leq q^h - c_{m,h} q^{h-2^{1-h}-4h/2^m}. \tag{18}$$

Now fix $\varepsilon > 0$. Choose m so large that $4h/2^m < \varepsilon/2$. Since $c_{m,h}$ is fixed and positive, for all sufficiently large q we also have $c_{m,h} \geq q^{-\varepsilon/2}$. The removed term in (18) is then at least $q^{h-2^{1-h}-\varepsilon}$, which proves (3). $\square$

Remark 7.1. The construction has two additional structural features: the removed set P is a product set, and the assigned direction of L_p depends only on the last coordinate of p. It is natural to ask whether the exponent $h - 2^{1-h}$ is optimal under these two restrictions.

8 Extending to prime powers

We start w

Let $\mathrm{IM}_*(h, q)$ denote the size of the largest induced matching $(p, s_p)_{p \in P}$ inside $\mathbb{F}_q^h$ where each slope s_p has non-zero h-coordinate. Note in particular that then $\mathrm{IM}_*(h, q) \geq (1/h)\,\mathrm{IM}(h, q)$ (by pigeonhole).

Let us now recover the following relationship.

Lemma 8.1. *Let q_0 be a prime power, and $q = q_0^s$. Then for $h = k(h_0 - 1) + 1$, we have*

$$\frac{\mathrm{IM}_*(k(h_0-1)+1, q)}{q^h} \geq q^{-1/2^k} \left(\frac{\mathrm{IM}_*(h_0, q_0)}{q_0^{h_0}}\right)^{(1-2^{-k})s}.$$

Proof. Let $f \in \mathbb{F}_{q_0}[T]$ be irreducible of degree s. We have that $\mathbb{F}_q \cong \mathbb{F}_{q_0}[T]/(f)$. Thus each $a \in \mathbb{F}_q$ has a unique representation as $\sum_{i=0}^{s-1} c_i T^i + (f)$. Let $\pi_i : \mathbb{F}_q \to \mathbb{F}_{q_0}$ map a to c_i, and note this map is $\mathbb{F}_{q_0}$-linear.

Now, let $i_0 := 0$ and for $t = 1, \dots, k$, let $i_t := i_{t-1} + \lceil (s - i_{t-1})/2 \rceil - 1$. Next, we define the discrete intervals $I_t := \{i_{t-1}, \dots, i_t - 1\}$. Note that $I := I_1 \sqcup \dots \sqcup I_k = \{0, 1, \dots, i_k\}$; and for each $i \in I$, there is a unique t so that $i \in I_t$, we set $i' := 2i - i_{t-1}$, and note that $i' < s$ for all $i \in I$.

Now, fix some induced matching $(p, s_p)_{p \in P_0}$ in $\mathbb{F}_{q_0}^{h_0}$. For $p \in P_0$, we may express $s_p = (\sigma_p, 1)$. Let us identify $\vec{a} \in \mathbb{F}_q^h$ with $(a^{(1)}, \dots, a^{(k)}, z) \in (\mathbb{F}_q^{h_0 - 1})^k \times \mathbb{F}_q$. Now, we design our point set $P \subset \mathbb{F}_q^h$. For $i \notin I$, we require $\pi_i(z) = 0$. For $t \in [k], i \in I_t$, we pick some $u_i \in \mathbb{F}_{q_0}^{h-1}, v_i \in \mathbb{F}_{q_0}$ so that $(u_i, v_i) \in P_0$, and require

$$\pi_i(z) = v_i, \quad \pi_{i'}(a^{(t)}) = u_i + \sum_{j=i'+1}^{i_t - 1} \pi_{2i-j}(z) \cdot \sigma_{(u_j, v_j)}.$$

Note that $|P|/q^h = q_0^{|I|-s} \left(\frac{|P_0|}{q_0}\right)^{|I|}$, thus showing P is an induced matching (with all slopes having non-zero h-coordinate) shall prove the claim.

It remains to define the slopes. Given $\vec{a}$, we assign $(a^{(1)}, \dots, a^{(k)}, z) \in P$ the slope $\vec{s} = (s^{(1)}, \dots, s^{(k)}, 1 + (f))$, where for $t \in [k], i \in I_t$, we set $\pi_{i'}(s^{(t)}) = \sigma_{(\pi_{i'}(a^{(t)}), \pi_i(z))}$, everywhere else, we assign the coefficient 0.

Okay, so now suppose $p' - p \in \langle s_p \rangle$. Suppose that $p = (a^{(1)}, \dots, a^{(k)}, z)$ is associated with $((u_i, v_i))_{i \in I}$ while $p' = (a'^{(1)}, \dots, a'^{(k)}, z)$ is associated with $((u_i', v_i'))_{i \in I}$. We must have that $z' - z = \lambda \neq 0$ (as the final coordinate of s_p has slope 1), thus there is a maximal i where $(u_i, v_i) \neq (u_i', v_i')$ (since $\pi_i(z' - z) = v_i' - v_i$).

We must have $i \in I_t$ for some t; we have

$$\pi_{i'}(a'^{(t)} - a^{(t)}) = u_i' - u_i + \sum_{j=i+1}^{i_t - 1} \pi_{2i-j}(z' - z) \cdot \sigma_{(u_j, v_j)}$$

(using that $(u_j, v_j) = (u_j', v_j')$ for all j in the sum, by maximality). At the same time

$$\pi_{i'}((z'-z) \cdot s_p^{(t)}) = \sum_{j \in I_t} \pi_{2i-j}(z'-z) \cdot \sigma_{(u_j, v_j)} = \sum_{j=i}^{i_t - 1} \pi_{2i-j}(z'-z) \cdot \sigma_{(u_j, v_j)} = \sigma_{(u_i, v_i)}(v_i' - v_i) + \sum_{j=i+1}^{i_t - 1} \pi_{2i-j}(z'-z) \cdot \sigma_{(u_j, v_j)}$$

(using that the terms for $j < i$ were all zero by maximality of i). Thus comparing the two equations we exactly get $u_i' = u_i + (v_i' - v_i) \cdot \sigma_{(u_i, v_i)}$, contradicting that $(u_i, v_i) \neq (u_i', v_i')$ (as P_0 was an induced matching). $\square$

Great, thus for fixed h, for all sufficiently large p we have $\mathrm{IM}(h_0, p) \geq p^{h - 2^{-h_0/3}}$ for all $h_0 \leq h$, whence taking $\mathrm{IM}(h, p^s) \geq p^{(h - \exp(-\Omega(h^{1/2})))s}$ (taking $k, h_0 = \Theta(h^{1/2})$). Meanwhile for fixed odd p, we have $\mathrm{IM}(h, p) \geq p^h - (p-1)((p+1)/2)^h = p^{h - O_p((1/2 + 1/2p)^h)}$, thus